

The Complete Memory Approximation of the 3D Euler's Equations

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1 Fourier Form of Euler's Equations

The three dimensional Euler's equations are given by:

$$\mathbf{u}_t + \mathbf{u} \cdot \nabla \mathbf{u} = -\nabla p, \quad \nabla \cdot \mathbf{u} = 0 \quad (1)$$

where $\mathbf{u}(x, t)$ is a three-dimensional vector periodic in all three dimensions and p is the pressure. Let the initial condition be $\mathbf{u}(x, 0) = \mathbf{u}^0$. We will write this equation in vector form in order to keep track of terms more carefully. Let u^x be the component of u in the x dimension (and similarly for y and z):

$$\begin{bmatrix} \frac{\partial u^x}{\partial t} \\ \frac{\partial u^y}{\partial t} \\ \frac{\partial u^z}{\partial t} \end{bmatrix} + \begin{bmatrix} u^x \frac{\partial u^x}{\partial x} + u^y \frac{\partial u^x}{\partial y} + u^z \frac{\partial u^x}{\partial z} \\ u^x \frac{\partial u^y}{\partial x} + u^y \frac{\partial u^y}{\partial y} + u^z \frac{\partial u^y}{\partial z} \\ u^x \frac{\partial u^z}{\partial x} + u^y \frac{\partial u^z}{\partial y} + u^z \frac{\partial u^z}{\partial z} \end{bmatrix} = - \begin{bmatrix} \frac{\partial p}{\partial x} \\ \frac{\partial p}{\partial y} \\ \frac{\partial p}{\partial z} \end{bmatrix}. \quad (2)$$

In order to specify the pressure p , we will apply $\nabla \cdot$ to both sides of the equation and use $\nabla \cdot \mathbf{u} = 0$.

$$\begin{aligned} \nabla \cdot \begin{bmatrix} \frac{\partial u^x}{\partial t} \\ \frac{\partial u^y}{\partial t} \\ \frac{\partial u^z}{\partial t} \end{bmatrix} + \nabla \cdot \begin{bmatrix} u^x \frac{\partial u^x}{\partial x} + u^y \frac{\partial u^x}{\partial y} + u^z \frac{\partial u^x}{\partial z} \\ u^x \frac{\partial u^y}{\partial x} + u^y \frac{\partial u^y}{\partial y} + u^z \frac{\partial u^y}{\partial z} \\ u^x \frac{\partial u^z}{\partial x} + u^y \frac{\partial u^z}{\partial y} + u^z \frac{\partial u^z}{\partial z} \end{bmatrix} &= -\nabla \cdot \begin{bmatrix} \frac{\partial p}{\partial x} \\ \frac{\partial p}{\partial y} \\ \frac{\partial p}{\partial z} \end{bmatrix} \\ \frac{\partial u^x}{\partial x} \frac{\partial u^x}{\partial x} + u^x \frac{\partial^2 u^x}{\partial x^2} + \frac{\partial u^y}{\partial x} \frac{\partial u^x}{\partial y} + u^y \frac{\partial^2 u^x}{\partial x \partial y} + \frac{\partial u^z}{\partial x} \frac{\partial u^x}{\partial z} + u^z \frac{\partial^2 u^x}{\partial x \partial z} \\ + \frac{\partial u^x}{\partial y} \frac{\partial u^y}{\partial x} + u^x \frac{\partial^2 u^y}{\partial x \partial y} + \frac{\partial u^y}{\partial y} \frac{\partial u^y}{\partial y} + u^y \frac{\partial^2 u^y}{\partial y^2} + \frac{\partial u^z}{\partial y} \frac{\partial u^y}{\partial z} + u^z \frac{\partial^2 u^y}{\partial y \partial z} \\ + \frac{\partial u^x}{\partial z} \frac{\partial u^z}{\partial x} + u^x \frac{\partial^2 u^z}{\partial x \partial z} + \frac{\partial u^y}{\partial z} \frac{\partial u^z}{\partial y} + u^y \frac{\partial^2 u^z}{\partial y \partial z} + \frac{\partial u^z}{\partial z} \frac{\partial u^z}{\partial z} + u^z \frac{\partial^2 u^z}{\partial z^2} &= - \left[\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial x^y 2} + \frac{\partial^2 p}{\partial z^2} \right]. \end{aligned}$$

Note that:

$$u^x \frac{\partial^2 u^x}{\partial x^2} + u^x \frac{\partial^2 u^y}{\partial x \partial y} + u^x \frac{\partial^2 u^z}{\partial x \partial z} = u^x \frac{\partial}{\partial x} [\nabla \cdot \mathbf{u}] = 0$$

and similiary for the $u^y \frac{\partial}{\partial y}$ and $u^z \frac{\partial}{\partial z}$ terms. Thus:

$$\begin{aligned} & \frac{\partial u^x}{\partial x} \frac{\partial u^x}{\partial x} + \frac{\partial u^y}{\partial x} \frac{\partial u^x}{\partial y} + \frac{\partial u^z}{\partial x} \frac{\partial u^x}{\partial z} \\ & + \frac{\partial u^x}{\partial y} \frac{\partial u^y}{\partial x} + \frac{\partial u^y}{\partial y} \frac{\partial u^y}{\partial y} + \frac{\partial u^z}{\partial y} \frac{\partial u^y}{\partial z} \\ & + \frac{\partial u^x}{\partial z} \frac{\partial u^z}{\partial x} + \frac{\partial u^y}{\partial z} \frac{\partial u^z}{\partial y} + \frac{\partial u^z}{\partial z} \frac{\partial u^z}{\partial z} = - \left[\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial x^2 \partial y^2} + \frac{\partial^2 p}{\partial z^2} \right] \end{aligned}$$

Because $\mathbf{u}$ is periodic in all three dimensions, we will write it as Fourier series:

$$\mathbf{u}(x, t) = \sum_{\mathbf{k}} \mathbf{u}(t) e^{i\mathbf{k} \cdot \mathbf{x}} \quad (3)$$

where $\mathbf{k}$ is a three-dimensional wavevector, and the sum is over all possible integer-valued wavevectors. Also let $p = \sum_{\mathbf{k}} p_{\mathbf{k}}(t) e^{i\mathbf{k} \cdot \mathbf{x}}$. We will be able to find a compact expression for $p_{\mathbf{k}}$. Consider first the right hand side:

$$\begin{aligned} - \left[\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + \frac{\partial^2 p}{\partial z^2} \right] &= - \left[\sum_{\mathbf{k}} -k_x^2 p_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{x}} + \sum_{\mathbf{k}} -k_y^2 p_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{x}} + \sum_{\mathbf{k}} -k_z^2 p_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{x}} \right] \\ &= \sum_{\mathbf{k}} |\mathbf{k}|^2 p_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{x}}. \end{aligned}$$

Now, the left hand side:

$$\begin{aligned}
& \frac{\partial u^x}{\partial x} \frac{\partial u^x}{\partial x} + \frac{\partial u^y}{\partial x} \frac{\partial u^x}{\partial y} + \frac{\partial u^z}{\partial x} \frac{\partial u^x}{\partial z} \\
& + \frac{\partial u^x}{\partial y} \frac{\partial u^y}{\partial x} + \frac{\partial u^y}{\partial y} \frac{\partial u^y}{\partial y} + \frac{\partial u^z}{\partial y} \frac{\partial u^y}{\partial z} \\
& + \frac{\partial u^x}{\partial z} \frac{\partial u^z}{\partial x} + \frac{\partial u^y}{\partial z} \frac{\partial u^z}{\partial y} + \frac{\partial u^z}{\partial z} \frac{\partial u^z}{\partial z} \\
& = \left(\sum_{\mathbf{p}} i p_x u_{\mathbf{p}}^x e^{i\mathbf{p} \cdot \mathbf{x}} \right) \left(\sum_{\mathbf{q}} i q_x u_{\mathbf{q}}^x e^{i\mathbf{q} \cdot \mathbf{x}} \right) + \left(\sum_{\mathbf{p}} i p_x u_{\mathbf{p}}^y e^{i\mathbf{p} \cdot \mathbf{x}} \right) \left(\sum_{\mathbf{q}} i q_y u_{\mathbf{q}}^x e^{i\mathbf{q} \cdot \mathbf{x}} \right) \\
& + \left(\sum_{\mathbf{p}} i p_x u_{\mathbf{p}}^z e^{i\mathbf{p} \cdot \mathbf{x}} \right) \left(\sum_{\mathbf{q}} i q_z u_{\mathbf{q}}^x e^{i\mathbf{q} \cdot \mathbf{x}} \right) + \left(\sum_{\mathbf{p}} i p_y u_{\mathbf{p}}^x e^{i\mathbf{p} \cdot \mathbf{x}} \right) \left(\sum_{\mathbf{q}} i q_x u_{\mathbf{q}}^y e^{i\mathbf{q} \cdot \mathbf{x}} \right) \\
& + \left(\sum_{\mathbf{p}} i p_y u_{\mathbf{p}}^y e^{i\mathbf{p} \cdot \mathbf{x}} \right) \left(\sum_{\mathbf{q}} i q_y u_{\mathbf{q}}^y e^{i\mathbf{q} \cdot \mathbf{x}} \right) + \left(\sum_{\mathbf{p}} i p_y u_{\mathbf{p}}^z e^{i\mathbf{p} \cdot \mathbf{x}} \right) \left(\sum_{\mathbf{q}} i q_z u_{\mathbf{q}}^y e^{i\mathbf{q} \cdot \mathbf{x}} \right) \\
& + \left(\sum_{\mathbf{p}} i p_z u_{\mathbf{p}}^x e^{i\mathbf{p} \cdot \mathbf{x}} \right) \left(\sum_{\mathbf{q}} i q_x u_{\mathbf{q}}^z e^{i\mathbf{q} \cdot \mathbf{x}} \right) + \left(\sum_{\mathbf{p}} i p_z u_{\mathbf{p}}^y e^{i\mathbf{p} \cdot \mathbf{x}} \right) \left(\sum_{\mathbf{q}} i q_y u_{\mathbf{q}}^z e^{i\mathbf{q} \cdot \mathbf{x}} \right) \\
& + \left(\sum_{\mathbf{p}} i p_z u_{\mathbf{p}}^z e^{i\mathbf{p} \cdot \mathbf{x}} \right) \left(\sum_{\mathbf{q}} i q_z u_{\mathbf{q}}^z e^{i\mathbf{q} \cdot \mathbf{x}} \right) \\
& = - \sum_{\mathbf{k}} \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} \left[p_x q_x u_{\mathbf{p}}^x u_{\mathbf{q}}^x + p_x q_y u_{\mathbf{p}}^y u_{\mathbf{q}}^x + p_x q_z u_{\mathbf{p}}^z u_{\mathbf{q}}^x \right. \\
& \quad \left. + p_y q_x u_{\mathbf{p}}^x u_{\mathbf{q}}^y + p_y q_y u_{\mathbf{p}}^y u_{\mathbf{q}}^y + p_y q_z u_{\mathbf{p}}^z u_{\mathbf{q}}^y \right. \\
& \quad \left. + p_z q_x u_{\mathbf{p}}^x u_{\mathbf{q}}^z + p_z q_y u_{\mathbf{p}}^y u_{\mathbf{q}}^z + p_z q_z u_{\mathbf{p}}^z u_{\mathbf{q}}^z \right] e^{i\mathbf{k} \cdot \mathbf{x}} \\
& = - \sum_{\mathbf{k}} \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{p} \cdot \mathbf{u}_{\mathbf{q}})(\mathbf{q} \cdot \mathbf{u}_{\mathbf{p}}) e^{i\mathbf{k} \cdot \mathbf{x}}.
\end{aligned}$$

Now observe that:

$$\begin{aligned}
\nabla \cdot \mathbf{u} = 0 & \implies 0 = \sum_{\mathbf{k}} [i k_x u_{\mathbf{k}}^x + i k_y u_{\mathbf{k}}^y + i k_z u_{\mathbf{k}}^z] e^{i\mathbf{k} \cdot \mathbf{x}} \\
0 & = i \sum_{\mathbf{k}} (\mathbf{k} \cdot \mathbf{u}_{\mathbf{k}}) e^{i\mathbf{k} \cdot \mathbf{x}}
\end{aligned}$$

Multiplying both sides by $e^{-i\mathbf{j} \cdot \mathbf{x}}$ and integrating yields:

$$0 = i(\mathbf{j} \cdot \mathbf{u}_{\mathbf{j}})$$

so we know that $\mathbf{k} \cdot \mathbf{u}_{\mathbf{k}} = 0$. Thus,

$$\mathbf{p} \cdot \mathbf{u}_{\mathbf{q}} = \mathbf{p} \cdot \mathbf{u}_{\mathbf{q}} + 0 = \mathbf{p} \cdot \mathbf{u}_{\mathbf{q}} + \mathbf{q} \cdot \mathbf{u}_{\mathbf{q}} = \mathbf{k} \cdot \mathbf{u}_{\mathbf{q}},$$

and similarly for the other dot product. Thus, the left hand side we had been simplifying is:

$$- \sum_{\mathbf{k}} \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{k} \cdot \mathbf{u}_{\mathbf{q}})(\mathbf{k} \cdot \mathbf{u}_{\mathbf{p}}) e^{i\mathbf{k} \cdot \mathbf{x}}$$

Multiplying the left and right hand sides by $e^{-i\mathbf{j}\cdot\mathbf{x}}$ leaves:

$$|\mathbf{j}|^2 p_{\mathbf{j}} = - \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{j} \cdot \mathbf{u}_{\mathbf{q}})(\mathbf{j} \cdot \mathbf{u}_{\mathbf{p}}) \implies p_{\mathbf{j}} = -\frac{1}{|\mathbf{j}|^2} \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{j} \cdot \mathbf{u}_{\mathbf{q}})(\mathbf{j} \cdot \mathbf{u}_{\mathbf{p}}).$$

We now return to the original equation, knowing we can plug this in for $p_{\mathbf{k}}$ later on. We write each term as a Fourier sum:

$$\begin{bmatrix} \frac{\partial u^x}{\partial t} \\ \frac{\partial u^y}{\partial t} \\ \frac{\partial u^z}{\partial t} \end{bmatrix} = \sum_{\mathbf{k}} \frac{d\mathbf{u}_{\mathbf{k}}}{dt} e^{i\mathbf{k}\cdot\mathbf{x}}$$

$$\begin{aligned} \begin{bmatrix} u^x \frac{\partial u^x}{\partial x} + u^y \frac{\partial u^x}{\partial y} + u^z \frac{\partial u^x}{\partial z} \\ u^x \frac{\partial u^y}{\partial x} + u^y \frac{\partial u^y}{\partial y} + u^z \frac{\partial u^y}{\partial z} \\ u^x \frac{\partial u^z}{\partial x} + u^y \frac{\partial u^z}{\partial y} + u^z \frac{\partial u^z}{\partial z} \end{bmatrix} &= \sum_{\mathbf{k}} \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} \begin{bmatrix} i q_x u_{\mathbf{p}}^x u_{\mathbf{q}}^x + i q_y u_{\mathbf{p}}^y u_{\mathbf{q}}^x + i q_z u_{\mathbf{p}}^z u_{\mathbf{q}}^x \\ i q_x u_{\mathbf{p}}^x u_{\mathbf{q}}^y + i q_y u_{\mathbf{p}}^y u_{\mathbf{q}}^y + i q_z u_{\mathbf{p}}^z u_{\mathbf{q}}^y \\ i q_x u_{\mathbf{p}}^x u_{\mathbf{q}}^z + i q_y u_{\mathbf{p}}^y u_{\mathbf{q}}^z + i q_z u_{\mathbf{p}}^z u_{\mathbf{q}}^z \end{bmatrix} e^{i\mathbf{k}\cdot\mathbf{x}} \\ &= i \sum_{\mathbf{k}} \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{q} \cdot \mathbf{u}_{\mathbf{p}}) \mathbf{u}_{\mathbf{q}} e^{i\mathbf{k}\cdot\mathbf{x}} \\ &= i \sum_{\mathbf{k}} \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{k} \cdot \mathbf{u}_{\mathbf{p}}) \mathbf{u}_{\mathbf{q}} e^{i\mathbf{k}\cdot\mathbf{x}} \end{aligned}$$

$$-\begin{bmatrix} \frac{\partial p}{\partial x} \\ \frac{\partial p}{\partial y} \\ \frac{\partial p}{\partial z} \end{bmatrix} = - \sum_{\mathbf{k}} \begin{bmatrix} i k_x p_{\mathbf{k}} \\ i k_y p_{\mathbf{k}} \\ i k_z p_{\mathbf{k}} \end{bmatrix} e^{i\mathbf{k}\cdot\mathbf{x}} = -i \sum_{\mathbf{k}} \mathbf{k} p_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} = i \sum_{\mathbf{k}} \frac{\mathbf{k}}{|\mathbf{k}|^2} \left(\sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{k} \cdot \mathbf{u}_{\mathbf{p}})(\mathbf{k} \cdot \mathbf{u}_{\mathbf{q}}) \right) e^{i\mathbf{k}\cdot\mathbf{x}}$$

Multiplying each term by $e^{-i\mathbf{j}\cdot\mathbf{x}}$ and integrating yields this equation for the evolution of a Fourier mode:

$$\frac{d\mathbf{u}_{\mathbf{k}}}{dt} = -i \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{k} \cdot \mathbf{u}_{\mathbf{p}}) \mathbf{u}_{\mathbf{q}} - \frac{\mathbf{k}}{|\mathbf{k}|^2} (\mathbf{k} \cdot \mathbf{u}_{\mathbf{p}})(\mathbf{k} \cdot \mathbf{u}_{\mathbf{q}}) \quad (4)$$

Written in a more compact form,

$$\frac{d\mathbf{u}_{\mathbf{k}}}{dt} = -i \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} \mathbf{k} \cdot \mathbf{u}_{\mathbf{p}} A_{\mathbf{k}} \mathbf{u}_{\mathbf{q}} \quad (5)$$

where

$$A_{\mathbf{k}} = I - \frac{\mathbf{k}\mathbf{k}^T}{|\mathbf{k}|^2}. \quad (6)$$

2 Implementation

Now we should think about how to implement this into Matlab. For a given state vector $\mathbf{u}$, the derivative of element k is given by (5). We will evolve the

system using the Runge-Kutta-Fehlberg method with adaptive stepsize using the `ode45` function.

This requires us to be able to evaluate the right hand side efficiently. Let's take a closer look at how we can do that. We first consider, in real space, the quantity $\mathbf{u}\mathbf{u}^T$:

$$\mathbf{u}\mathbf{u}^T = \begin{bmatrix} u^x u^x & u^x u^y & u^x u^z \\ u^y u^x & u^y u^y & u^y u^z \\ u^z u^x & u^z u^y & u^z u^z \end{bmatrix} = \sum_{\mathbf{k}} \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} \begin{bmatrix} u_{\mathbf{p}}^x u_{\mathbf{q}}^x & u_{\mathbf{p}}^y u_{\mathbf{q}}^x & u_{\mathbf{p}}^z u_{\mathbf{q}}^x \\ u_{\mathbf{p}}^x u_{\mathbf{q}}^y & u_{\mathbf{p}}^y u_{\mathbf{q}}^y & u_{\mathbf{p}}^z u_{\mathbf{q}}^y \\ u_{\mathbf{p}}^x u_{\mathbf{q}}^z & u_{\mathbf{p}}^y u_{\mathbf{q}}^z & u_{\mathbf{p}}^z u_{\mathbf{q}}^z \end{bmatrix} e^{i\mathbf{k}\cdot\mathbf{x}}.$$

The fast Fourier transform of this is found by multiplying by $e^{-i\mathbf{j}\cdot\mathbf{x}}$ and integration. Thus,

$$[FFT(\mathbf{u}\mathbf{u}^T)]_{\mathbf{k}} = \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} \begin{bmatrix} u_{\mathbf{p}}^x u_{\mathbf{q}}^x & u_{\mathbf{p}}^y u_{\mathbf{q}}^x & u_{\mathbf{p}}^z u_{\mathbf{q}}^x \\ u_{\mathbf{p}}^x u_{\mathbf{q}}^y & u_{\mathbf{p}}^y u_{\mathbf{q}}^y & u_{\mathbf{p}}^z u_{\mathbf{q}}^y \\ u_{\mathbf{p}}^x u_{\mathbf{q}}^z & u_{\mathbf{p}}^y u_{\mathbf{q}}^z & u_{\mathbf{p}}^z u_{\mathbf{q}}^z \end{bmatrix}.$$

Observe the two terms in the right hand side of (5). First is,

$$\begin{aligned} -i \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{k} \cdot \mathbf{u}_{\mathbf{p}}) \mathbf{u}_{\mathbf{q}} &= -i \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} \mathbf{u}_{\mathbf{q}} \mathbf{u}_{\mathbf{p}}^T \mathbf{k} \\ &= -i \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} \begin{bmatrix} u_{\mathbf{p}}^x u_{\mathbf{q}}^x & u_{\mathbf{p}}^y u_{\mathbf{q}}^x & u_{\mathbf{p}}^z u_{\mathbf{q}}^x \\ u_{\mathbf{p}}^x u_{\mathbf{q}}^y & u_{\mathbf{p}}^y u_{\mathbf{q}}^y & u_{\mathbf{p}}^z u_{\mathbf{q}}^y \\ u_{\mathbf{p}}^x u_{\mathbf{q}}^z & u_{\mathbf{p}}^y u_{\mathbf{q}}^z & u_{\mathbf{p}}^z u_{\mathbf{q}}^z \end{bmatrix} \mathbf{k} \\ &= -i [FFT(\mathbf{u}\mathbf{u}^T)]_{\mathbf{k}} \mathbf{k}. \end{aligned}$$

Thus, it can be computed easily by transforming into real space, multiplying the state vectors, and returning to Fourier space. Similarly, the second term is:

$$\begin{aligned} i \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} \mathbf{k} \cdot \mathbf{u}_{\mathbf{p}} \frac{\mathbf{k}\mathbf{k}^T}{|\mathbf{k}|^2} \mathbf{u}_{\mathbf{q}} &= \frac{i\mathbf{k}}{|\mathbf{k}|^2} \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} (\mathbf{k} \cdot \mathbf{u}_{\mathbf{p}}) (\mathbf{k} \cdot \mathbf{u}_{\mathbf{q}}) \\ &= \frac{i\mathbf{k}}{|\mathbf{k}|^2} \sum_{\mathbf{p}+\mathbf{q}=\mathbf{k}} \begin{bmatrix} k_x k_x u_{\mathbf{p}}^x u_{\mathbf{q}}^x + k_x k_y u_{\mathbf{p}}^y u_{\mathbf{q}}^x + k_x k_z u_{\mathbf{p}}^z u_{\mathbf{q}}^x \\ k_y k_x u_{\mathbf{p}}^x u_{\mathbf{q}}^y + k_y k_y u_{\mathbf{p}}^y u_{\mathbf{q}}^y + k_y k_z u_{\mathbf{p}}^z u_{\mathbf{q}}^y \\ k_z k_x u_{\mathbf{p}}^x u_{\mathbf{q}}^z + k_z k_y u_{\mathbf{p}}^y u_{\mathbf{q}}^z + k_z k_z u_{\mathbf{p}}^z u_{\mathbf{q}}^z \end{bmatrix} \\ &= \frac{i\mathbf{k}}{|\mathbf{k}|^2} (\mathbf{k}^T [FFT(\mathbf{u}\mathbf{u}^T)]_{\mathbf{k}} \mathbf{k}). \end{aligned}$$

Thus, we can also compute the other part of the right hand side using the same real space calculation.

3 Reduced Models

Unlike the KdV case, the reduced models for Euler are simpler to compute. This is largely due to the fact that the RHS has only one term, so the models

do not telescope out in a challenging fashion. We do have to note that the convolution is no longer symmetric. With the benefit of hindsight, we can define our functions ahead of time to make our life easier when applying repeated differential operators.

Let $\mathbf{u}(t)$ be the solution to Euler's equations. Consider the Fourier components $\mathbf{u}_{\mathbf{k}}(t)$, where $\mathbf{k} \in F \cup G$. Let F be the set of resolved modes. That is $F = \{\mathbf{k} \in [-N, N] \times [-N, N] \times [-N, N]\}$. Let $F \cup G = \{\mathbf{k} \in [-M, M] \times [-M, M] \times [-M, M]\}$. Define $\hat{\mathbf{u}} = \{\mathbf{u}_{\mathbf{k}} \mid \mathbf{k} \in F\}$ and $\tilde{\mathbf{u}} = \{\mathbf{u}_{\mathbf{k}} \mid \mathbf{k} \in G\}$.

Following the Mori-Zwanzig formalism, we define

$$\mathcal{L} = \sum_{\mathbf{k} \in F \cup G} \begin{bmatrix} R_{\mathbf{k}}^x(\mathbf{u}^0) \frac{\partial}{\partial u_{\mathbf{k}}^{0,x}} & 0 & 0 \\ 0 & R_{\mathbf{k}}^y(\mathbf{u}^0) \frac{\partial}{\partial u_{\mathbf{k}}^{0,y}} & 0 \\ 0 & 0 & R_{\mathbf{k}}^z(\mathbf{u}^0) \frac{\partial}{\partial u_{\mathbf{k}}^{0,z}} \end{bmatrix}$$

where $R_{\mathbf{k}}^i$ is the component of $\mathbf{R}_{\mathbf{k}}$ in the i th direction and $u_{\mathbf{k}}^{0,i}$ is the component of the initial condition of $u_{\mathbf{k}}$ in the i th direction. Observe that:

$$\mathcal{L}\mathbf{u}_{\mathbf{k}}^0 = \mathbf{R}_{\mathbf{k}}(\mathbf{u}^0)$$

as desired. Therefore, we can see

$$\mathcal{L}\mathbf{u}_{\mathbf{k}}^0 = -i \sum_{\substack{\mathbf{p}+\mathbf{q}=\mathbf{k} \\ \mathbf{p}, \mathbf{q}, \mathbf{k} \in F \cup G}} \mathbf{k} \cdot \mathbf{u}_{\mathbf{p}}^0 A_{\mathbf{k}} \mathbf{u}_{\mathbf{q}}^0$$

where $A_{\mathbf{k}} = I - \frac{\mathbf{k}\mathbf{k}^T}{|\mathbf{k}|^2}$.

Let's define the convolution $\mathbf{C}$ of two vectors. Let $\mathbf{C}_{\mathbf{k}}$ be the component of $\mathbf{C}$ corresponding to the wavevector $\mathbf{k}$:

$$\mathbf{C}_{\mathbf{k}}(\mathbf{v}, \mathbf{w}) = -i \sum_{\substack{\mathbf{p}+\mathbf{q}=\mathbf{k} \\ \mathbf{p}, \mathbf{q} \in F \cup G}} \mathbf{k} \cdot \mathbf{v}_{\mathbf{p}} A_{\mathbf{k}} \mathbf{w}_{\mathbf{q}} \quad (7)$$

where $A_{\mathbf{k}} = I - \frac{\mathbf{k}\mathbf{k}^T}{|\mathbf{k}|^2}$. Note crucially that $A_{\mathbf{k}}$ has no dependence upon any $\mathbf{u}_{\mathbf{k}}$.

Under this definition, we can see that $\mathcal{L}\mathbf{u}_{\mathbf{k}}^0 = \mathbf{C}_{\mathbf{k}}(\mathbf{u}^0, \mathbf{u}^0)$, or under the Mori-Zwanzig formalism that

$$\frac{d\mathbf{u}_{\mathbf{k}}}{dt} = e^{t\mathcal{L}} \mathbf{C}_{\mathbf{k}}(\mathbf{u}^0, \mathbf{u}^0) = \mathbf{C}_{\mathbf{k}}(\mathbf{u}, \mathbf{u}).$$

. For this to be the exact full solution, we would need $M = \infty$. For our implementation, M will necessarily be the maximal mode we choose to retain.

We now define the projector P to be

$$Pf(\mathbf{u}) = Pf(\hat{\mathbf{u}}^0, \tilde{\mathbf{u}}^0) = f(\hat{\mathbf{u}}^0, 0).$$

That is, it simply sets all unresolved modes to zero. Let $Q = I - P$. Then we can begin constructing the terms of a complete memory approximation:

$$\frac{d\mathbf{u}_{\mathbf{k}}}{dt} = \sum_{i=0}^4 a_i(t) t^i \hat{\mathbf{R}}_{\mathbf{k}}^i(\mathbf{u}).$$

3.1 Markov Term

The Markov term is

$$\hat{\mathbf{R}}_{\mathbf{k}}^0(\mathbf{u}) = Pe^{t\mathcal{L}}P\mathcal{L}\mathbf{u}_{\mathbf{k}}^0.$$

Using the definitions above, we find:

$$\begin{aligned}\hat{\mathbf{R}}_{\mathbf{k}}^0(\mathbf{u}) &= Pe^{t\mathcal{L}}P\mathbf{C}_{\mathbf{k}}(\mathbf{u}^0, \mathbf{u}^0) \\ &= Pe^{t\mathcal{L}}\mathbf{C}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0) \\ &= \mathbf{C}_{\mathbf{k}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}).\end{aligned}$$

Here, we used the fact that P sets the unresolved modes to zero. Through a slight abuse of notation, consider $\hat{\mathbf{u}}^0$ to be the array of $\mathbf{u}$ where all $\mathbf{u}_{\mathbf{k}} = 0$ for $\mathbf{k} \in G$ (and similarly for $\hat{\mathbf{u}}$).

3.2 t -model Term

The t -model is

$$\hat{\mathbf{R}}_{\mathbf{k}}^1(\mathbf{u}) = Pe^{t\mathcal{L}}P\mathcal{L}Q\mathcal{L}\mathbf{u}_{\mathbf{k}}^0.$$

We can simplify this as:

$$\begin{aligned}\hat{\mathbf{R}}_{\mathbf{k}}^1(\mathbf{u}) &= Pe^{t\mathcal{L}}P\mathcal{L}Q\mathcal{L}\mathbf{u}_{\mathbf{k}}^0 \\ &= Pe^{t\mathcal{L}}P\mathcal{L}[\mathcal{L} - P\mathcal{L}]\mathbf{u}_{\mathbf{k}}^0 \\ &= Pe^{t\mathcal{L}}P\mathcal{L}[\mathbf{C}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{u}}^0) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{u}}^0, \hat{\mathbf{u}}^0) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{u}}^0, \tilde{\mathbf{u}}^0)]\end{aligned}$$

In order to proceed, we must investigate how $\mathcal{L}$ operates on a convolution sum:

$$\begin{aligned}\mathcal{L}\mathbf{C}_{\mathbf{k}}(\mathbf{v}, \mathbf{w}) &= \mathcal{L} \left(-i \sum_{\substack{\mathbf{p}+\mathbf{q}=\mathbf{k} \\ \mathbf{p}, \mathbf{q} \in F \cup G}} \mathbf{k} \cdot \mathbf{v}_{\mathbf{p}} A_{\mathbf{k}} \mathbf{w}_{\mathbf{q}} \right) \\ &= -i \sum_{\substack{\mathbf{p}+\mathbf{q}=\mathbf{k} \\ \mathbf{p}, \mathbf{q} \in F \cup G}} \mathbf{k} \cdot (\mathcal{L}\mathbf{v}_{\mathbf{p}}) A_{\mathbf{k}} \mathbf{w}_{\mathbf{q}} - i \sum_{\substack{\mathbf{p}+\mathbf{q}=\mathbf{k} \\ \mathbf{p}, \mathbf{q} \in F \cup G}} \mathbf{k} \cdot \mathbf{v}_{\mathbf{p}} A_{\mathbf{k}} (\mathcal{L}\mathbf{w}_{\mathbf{q}}) \\ &= \mathbf{C}_{\mathbf{k}}(\mathcal{L}\mathbf{v}, \mathbf{w}) + \mathbf{C}_{\mathbf{k}}(\mathbf{v}, \mathcal{L}\mathbf{w}).\end{aligned}$$

We used product rule. Now consider how $\mathcal{L}$ operates on $\hat{\mathbf{u}}^0$ and $\tilde{\mathbf{u}}^0$. Consider $\mathbf{p} \in F$ and $\mathbf{q} \in G$:

$$\begin{aligned}\mathcal{L}\hat{\mathbf{u}}_{\mathbf{p}}^0 &= -i \sum_{\substack{\mathbf{r}+\mathbf{s}=\mathbf{p} \\ \mathbf{r}, \mathbf{s} \in F \cup G, \mathbf{p} \in F}} \mathbf{p} \cdot \mathbf{u}_{\mathbf{r}}^0 A_{\mathbf{p}} \mathbf{u}_{\mathbf{s}}^0 = \mathbf{C}_{\mathbf{p}}(\mathbf{u}^0, \mathbf{u}^0) \\ \mathcal{L}\tilde{\mathbf{u}}_{\mathbf{q}}^0 &= -i \sum_{\substack{\mathbf{r}+\mathbf{s}=\mathbf{q} \\ \mathbf{r}, \mathbf{s} \in F \cup G, \mathbf{q} \in G}} \mathbf{p} \cdot \mathbf{u}_{\mathbf{r}}^0 A_{\mathbf{p}} \mathbf{u}_{\mathbf{s}}^0 = \mathbf{C}_{\mathbf{q}}(\mathbf{u}^0, \mathbf{u}^0).\end{aligned}$$

Define the following arrays. Let $\hat{\mathbf{C}}(\mathbf{v}, \mathbf{w})$ be the convolution of $\mathbf{v}$ and $\mathbf{w}$, but set all indices corresponding to $\mathbf{k} \in G$ to zero. Let $\tilde{\mathbf{C}}(\mathbf{v}, \mathbf{w})$ be the convolution of

$\mathbf{v}$ and $\mathbf{w}$, but set all indices corresponding to $\mathbf{k} \in F$ to zero. Thus, $\mathbf{C}(\mathbf{v}, \mathbf{w}) = \hat{\mathbf{C}}(\mathbf{v}, \mathbf{w}) + \tilde{\mathbf{C}}(\mathbf{v}, \mathbf{w})$. Observe that, under this definition (and based upon the componentwise derivations above), $\mathcal{L}\hat{\mathbf{u}}^0 = \hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0)$ and $\mathcal{L}\tilde{\mathbf{u}}^0 = \tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0)$. Thus,

$$\begin{aligned}\hat{\mathbf{R}}_{\mathbf{k}}^1(\mathbf{u}) &= Pe^{t\mathcal{L}}P\mathcal{L}[\mathbf{C}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{u}}^0) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{u}}^0, \hat{\mathbf{u}}^0) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{u}}^0, \tilde{\mathbf{u}}^0)] \\ &= Pe^{t\mathcal{L}}P[\mathbf{C}_{\mathbf{k}}(\hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{u}}^0) + \mathbf{C}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0)) \\ &\quad + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \hat{\mathbf{u}}^0) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{u}}^0, \hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0)) \\ &\quad + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{u}}^0) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{u}}^0, \tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0))].\end{aligned}$$

In order to apply the projector, observe that $P\mathbf{C}_{\mathbf{k}}(\mathbf{v}, \mathbf{w}) = \mathbf{C}_{\mathbf{k}}(P\mathbf{v}, P\mathbf{w})$. Note that $P\mathbf{u}^0 = \hat{\mathbf{u}}^0$, $P\hat{\mathbf{u}}^0 = \hat{\mathbf{u}}^0$, and $P\tilde{\mathbf{u}}^0 = 0$. Finally, note that $\mathbf{C}_{\mathbf{k}}(\mathbf{v}, 0) = \mathbf{C}_{\mathbf{k}}(0, \mathbf{w}) = 0$. This yields:

$$\begin{aligned}\hat{\mathbf{R}}_{\mathbf{k}}^1(\mathbf{u}) &= Pe^{t\mathcal{L}}P[\mathbf{C}_{\mathbf{k}}(\hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{u}}^0) + \mathbf{C}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0)) \\ &\quad + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \hat{\mathbf{u}}^0) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{u}}^0, \hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0)) \\ &\quad + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{u}}^0) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{u}}^0, \tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0))] \\ &= Pe^{t\mathcal{L}}[\mathbf{C}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0)) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0), \hat{\mathbf{u}}^0)] \\ &= \mathbf{C}_{\mathbf{k}}(\hat{\mathbf{u}}, \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \hat{\mathbf{u}}).\end{aligned}$$

This term involves convolutions of the same two terms in both permutations. It is useful to define a function:

$$\mathbf{D}(\mathbf{v}, \mathbf{w}) = \mathbf{C}(\mathbf{v}, \mathbf{w}) + \mathbf{C}(\mathbf{w}, \mathbf{v})$$

and the related functions $\mathbf{D}_{\mathbf{k}}$, $\hat{\mathbf{D}}$, and $\tilde{\mathbf{D}}$ defined in terms of the equivalent convolutions. With this notation, we can write the t -model term in a single expression:

$$\hat{\mathbf{R}}_{\mathbf{k}}^1(\mathbf{u}) = \mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}, \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))$$

3.3 t^2 -model Term

The set of rules of operation derived in the previous section will allow us to proceed to ever higher terms. The t^2 -model is:

$$\hat{\mathbf{R}}_{\mathbf{k}}^2(\mathbf{u}) = \frac{1}{2}Pe^{t\mathcal{L}}P\mathcal{L}[Q\mathcal{L} - P\mathcal{L}]Q\mathcal{L}\mathbf{u}_{\mathbf{k}}^0.$$

First note that, with our newly defined $\mathbf{D}$ function,

$$Q\mathcal{L}\mathbf{u}_{\mathbf{k}}^0 = \mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}, \tilde{\mathbf{u}}) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{u}}, \tilde{\mathbf{u}}).$$

We will also need to understand how $\mathcal{L}$ operates upon $\mathbf{D}$:

$$\begin{aligned}\mathcal{L}\mathbf{D}(\mathbf{v}, \mathbf{w}) &= \mathcal{L}\mathbf{C}(\mathbf{v}, \mathbf{w}) + \mathcal{L}\mathbf{C}(\mathbf{w}, \mathbf{v}) \\ &= \mathbf{C}(\mathcal{L}\mathbf{v}, \mathbf{w}) + \mathbf{C}(\mathbf{v}, \mathcal{L}\mathbf{w}) + \mathbf{C}(\mathcal{L}\mathbf{w}, \mathbf{v}) + \mathbf{C}(\mathbf{w}, \mathcal{L}\mathbf{v}) \\ &= \mathbf{D}(\mathcal{L}\mathbf{v}, \mathbf{w}) + \mathbf{D}(\mathbf{v}, \mathcal{L}\mathbf{w}).\end{aligned}$$

Thus, it operates in the same manner it did upon $\mathbf{C}$. The projector P similarly is applied to each term within the expression

$$P\mathbf{D}(\mathbf{v}, \mathbf{w}) = \mathbf{D}(P\mathbf{v}, P\mathbf{w}).$$

Starting from this, we derive an expression for the t^2 -term:

$$\begin{aligned}\hat{\mathbf{R}}_{\mathbf{k}}^2(\mathbf{u}) &= \frac{1}{2}Pe^{t\mathcal{L}}P\mathcal{L}(\mathcal{L} - 2P\mathcal{L})[\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{u}}^0) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{u}}^0, \tilde{\mathbf{u}}^0)] \\ &= \frac{1}{2}Pe^{t\mathcal{L}}P\mathcal{L}[\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{u}}^0) + \mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0)) \\ &\quad + \mathbf{D}_{\mathbf{k}}(\tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{u}}^0) - 2\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0))] \\ &= \frac{1}{2}Pe^{t\mathcal{L}}[\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{D}}(\tilde{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0) - \hat{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0), \hat{\mathbf{u}}^0)) + \mathbf{D}_{\mathbf{k}}(\tilde{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0), \tilde{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0))] \\ &= \frac{1}{2}\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}, \tilde{\mathbf{D}}(\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \hat{\mathbf{u}})) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})).\end{aligned}$$

3.4 t^3 -model Term

We will make use of the terms we have already computed in order to simplify our derivation of the third order term. Under the complete memory approximation, the third term is:

$$\hat{\mathbf{R}}_{\mathbf{k}}^3(\mathbf{u}) = \frac{1}{6}Pe^{t\mathcal{L}}P\mathcal{L}[PLPL - 2PLQL - 2QLPL + QLQL]Q\mathcal{L}\mathbf{u}_{\mathbf{k}}^0.$$

Once again, we rewrite it with no $Q\mathcal{L}$ terms.

$$\begin{aligned}\hat{\mathbf{R}}_{\mathbf{k}}^3(\mathbf{u}) &= \frac{1}{6}Pe^{t\mathcal{L}}P\mathcal{L}[PLPL - 2PLQL - 2QLPL + QLQL]Q\mathcal{L}\mathbf{u}_{\mathbf{k}}^0 \\ &= \frac{1}{6}Pe^{t\mathcal{L}}P\mathcal{L}[-3P\mathcal{L}(\mathcal{L} - 2P\mathcal{L}) - 3\mathcal{L}P\mathcal{L} + \mathcal{L}\mathcal{L}]Q\mathcal{L}\mathbf{u}_{\mathbf{k}}^0\end{aligned}$$

We recognize that we have already computed an expression for $P\mathcal{L}(\mathcal{L} - 2P\mathcal{L})Q\mathcal{L}\mathbf{u}_{\mathbf{k}}^0$ in the previous subsection. This leaves two additional terms to compute before simplifying and applying the final $P\mathcal{L}$:

$$\begin{aligned}\mathcal{L}P\mathcal{L}Q\mathcal{L}\mathbf{u}_{\mathbf{k}}^0 &= \mathcal{L}[\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0))] \\ &= \mathbf{D}_{\mathbf{k}}(\hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0)) + \mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{D}}(\hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \hat{\mathbf{u}}^0)) \\ \mathcal{L}\mathcal{L}Q\mathcal{L}\mathbf{u}_{\mathbf{k}}^0 &= \mathcal{L}\mathcal{L}[\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{u}}^0) + \mathbf{C}_{\mathbf{k}}(\tilde{\mathbf{u}}^0, \tilde{\mathbf{u}}^0)] \\ &= \mathcal{L}[\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{u}}^0) + \mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}^0, \tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0)) + \mathbf{D}_{\mathbf{k}}(\tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{u}}^0)] \\ &= \mathbf{D}_{\mathbf{k}}(\hat{\mathbf{D}}(\mathbf{C}(\mathbf{u}^0, \mathbf{u}^0), \mathbf{u}^0), \tilde{\mathbf{u}}^0) + 2\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0)) \\ &\quad + \mathbf{D}_{\mathbf{k}}(\mathbf{u}^0, \tilde{\mathbf{D}}(\mathbf{C}(\mathbf{u}^0, \mathbf{u}^0), \mathbf{u}^0)) + \mathbf{D}_{\mathbf{k}}(\tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0)).\end{aligned}$$

Combining these three computed terms with the correct coefficients yields:

$$\begin{aligned}
[-3P\mathcal{L}(\mathcal{L} - 2P\mathcal{L}) - 3\mathcal{L}P\mathcal{L} + \mathcal{L}\mathcal{L}]Q\mathcal{L}\mathbf{u}_k^0 = & \mathbf{D}_k(\hat{\mathbf{u}}^0, \tilde{\mathbf{D}}(-3\hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0) + 3\hat{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0) - 3\tilde{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0), \hat{\mathbf{u}}^0)) \\
& - 6\mathbf{C}_k(\tilde{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0), \tilde{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0)) - 3\mathbf{D}_k(\hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{C}}(\hat{\mathbf{u}}^0, \hat{\mathbf{u}}^0)) \\
& + \mathbf{D}_k(\hat{\mathbf{D}}(\mathbf{C}(\mathbf{u}^0, \mathbf{u}^0), \mathbf{u}^0), \tilde{\mathbf{u}}^0) + 2\mathbf{D}_k(\hat{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0)) \\
& + \mathbf{D}_k(\mathbf{u}^0, \tilde{\mathbf{D}}(\mathbf{C}(\mathbf{u}^0, \mathbf{u}^0), \mathbf{u}^0)) + 2\mathbf{C}_k(\tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0), \tilde{\mathbf{C}}(\mathbf{u}^0, \mathbf{u}^0)).
\end{aligned}$$

The t^3 -term is found once we apply $P\mathcal{L}$ to this expression, yielding:

$$\begin{aligned}
\hat{\mathbf{R}}_k^3(\mathbf{u}) = & \frac{1}{6}e^{t\mathcal{L}}P\mathcal{L}[-3P\mathcal{L}(\mathcal{L} - 2P\mathcal{L}) - 3\mathcal{L}P\mathcal{L} + \mathcal{L}\mathcal{L}]Q\mathcal{L}\mathbf{u}_k^0 \\
= & \frac{1}{6}[\mathbf{D}_k(\hat{\mathbf{u}}, \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 2\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 2\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
& + \tilde{\mathbf{D}}(\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) \\
& + 3\mathbf{D}_k(\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})))].
\end{aligned}$$

3.5 t^4 -model Term

Our final included model will be the t^4 -model. The derivation of these models is quite tedious. For this reason, we built a symbolic notebook to compute these

terms automatically. The result for the fourth order model is:

$$\begin{aligned}
\hat{\mathbf{R}}_{\mathbf{k}}^4(\mathbf{u}) &= \frac{1}{24} e^{t\mathcal{L}} P\mathcal{L}[PLPLPL - 3PLPLQL - 5PLQLPL + 3PLQLQL \\
&\quad - 3QLPLPL + 5QLPLQL + 3QLQLPL - QLQLQL] Q\mathcal{L}\mathbf{u}_{\mathbf{k}}^0 \\
&= -\frac{1}{24} [12\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})), \hat{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad + 3\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{D}}(\hat{\mathbf{u}}, \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}})), \hat{\mathbf{D}}(\hat{\mathbf{u}}, \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad + 12\mathbf{D}_{\mathbf{k}}(\mathbf{D}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})), \mathbf{D}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad + 3\mathbf{D}_{\mathbf{k}}(\mathbf{D}(\hat{\mathbf{u}}, \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}})), \mathbf{D}(\hat{\mathbf{u}}, \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad - \mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}, \tilde{\mathbf{D}}(\hat{\mathbf{u}}, 6\hat{\mathbf{D}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + 3\hat{\mathbf{D}}(\mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) \\
&\quad \quad - 6\tilde{\mathbf{D}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) - \tilde{\mathbf{D}}(\mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) \\
&\quad \quad + \hat{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 3\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, 3\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 5\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad \quad + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{D}}(\hat{\mathbf{u}}, 5\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 3\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, 3\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad \quad - 8\hat{\mathbf{D}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + 4\tilde{\mathbf{D}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad \quad + \tilde{\mathbf{D}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), 4\hat{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 2\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, 4\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 8\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad \quad + \tilde{\mathbf{D}}(\mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \hat{\mathbf{D}}(\hat{\mathbf{u}}, 3\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, 5\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 3\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})))) \\
&\quad - \mathbf{D}_{\mathbf{k}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), 24\tilde{\mathbf{C}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + 8\tilde{\mathbf{C}}(\mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) - 12\tilde{\mathbf{D}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) \\
&\quad \quad + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, 4\hat{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 2\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, -8\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) + 4\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad - \mathbf{D}_{\mathbf{k}}(\mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), -24\tilde{\mathbf{C}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) - 8\tilde{\mathbf{C}}(\mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + 12\tilde{\mathbf{D}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) \\
&\quad \quad + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{D}}(\hat{\mathbf{u}}, -4\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) + 8\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, 8\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 4\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})))) \\
&\quad - 12\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})), \hat{\mathbf{D}}(\hat{\mathbf{u}}, 3\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) + \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + 2\tilde{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad - \mathbf{D}_{\mathbf{k}}(\hat{\mathbf{D}}(\hat{\mathbf{u}}, -6\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) + 6\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})), \mathbf{D}(\hat{\mathbf{u}}, \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad - 12\mathbf{D}_{\mathbf{k}}(\mathbf{D}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})), \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \mathbf{C}(\hat{\mathbf{u}}, \hat{\mathbf{u}})))]
\end{aligned}$$

After much simplifying (of which I'm not especially confident), we get:

$$\begin{aligned}
\hat{\mathbf{R}}_{\mathbf{k}}^4(\mathbf{u}) &= -\frac{1}{24}e^{t\mathcal{L}}P\mathcal{L}[PLPLPL - 3PLPLQL - 5PLQLPL + 3PLQLQL \\
&\quad - 3QLPLPL + 5QLPLQL + 3QLQLPL - QLQLQL]Q\mathcal{L}\mathbf{u}_{\mathbf{k}}^0 \\
&= \frac{1}{24}[\mathbf{D}_{\mathbf{k}}(\hat{\mathbf{u}}, \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{D}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 2\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + 3\hat{\mathbf{D}}(\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) \\
&\quad + \tilde{\mathbf{D}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), 2\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 3\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) - \tilde{\mathbf{D}}(\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) \\
&\quad + \hat{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 3\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, 3\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 5\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{D}}(\hat{\mathbf{u}}, 5\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 3\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, 3\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad + \tilde{\mathbf{D}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \hat{\mathbf{D}}(\hat{\mathbf{u}}, 3\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 5\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 3\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad + \tilde{\mathbf{D}}(\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \hat{\mathbf{D}}(\hat{\mathbf{u}}, 3\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, 5\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 3\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad + 4\mathbf{D}_{\mathbf{k}}(\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \tilde{\mathbf{D}}(\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}), \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) \\
&\quad + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 2\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})) + \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 2\hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad + 3\mathbf{D}_{\mathbf{k}}(\tilde{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})), \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \hat{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}) - 2\tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}}))) \\
&\quad + 3\mathbf{D}_{\mathbf{k}}(\tilde{\mathbf{D}}(\hat{\mathbf{u}}, \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})), \tilde{\mathbf{D}}(\hat{\mathbf{u}}, \tilde{\mathbf{C}}(\hat{\mathbf{u}}, \hat{\mathbf{u}})))]
\end{aligned}$$